

Lab 9

NO late submission. Grade = 0 after the deadline.

Due: 13:50h 03/12/2025

Submission:

- **Lab#_YOURNAME.ipynb**
- Must include all data file (**.csv**,...) and **.ipynb** in your submission
- **.csv** and **.ipynb** must be in the **same** folder (pd.read_csv('data.csv'))

No use of LOCAL libraries, NO local directory for libraries

Question 1:

When calculating the **sample standard deviation** (or **sample variance**), why do we divide by **(N-1)** instead of **N**?

Question 2:

Use '**house_prices.csv**'.

The selected features are highlighted:

Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape
	LandContour	Utilities	LotConfig	LandSlope	NeighborhoodCondition1		
	Condition2	BldgType	HouseStyle	OverallQual	OverallCond	YearBuilt	
	YearRemodAdd	RoofStyle	RoofMatl	Exterior1st	Exterior2nd		
	MasVnrType	MasVnrArea	ExterQual	ExterCond	Foundation	BsmtQual	
	BsmtCond	BsmtExposure	BsmtFinType1	BsmtFinSF1	BsmtFinType2	BsmtFinSF2	
	BsmtUnfSF	TotalBsmtSF	Heating	HeatingQC	CentralAir	Electrical	
	1stFlrSF	2ndFlrSF	LowQualFinSF	GrLivArea	BsmtFullBath	BsmtHalfBath	
	FullBath	HalfBath	BedroomAbvGr	KitchenAbvGr	KitchenQual		
	TotRmsAbvGrd	Functional	Fireplaces	FireplaceQu	GarageType		
	GarageYrBlt	GarageFinish	GarageCars	GarageArea	GarageQual	GarageCond	
	PavedDrive	WoodDeckSF	OpenPorchSF	EnclosedPorch	3SsnPorch		
	ScreenPorch	PoolArea	PoolQC	Fence	MiscFeature	MiscVal	
	MoSold	YrSold	SaleType	SaleCondition	SalePrice		

- a) Compute the covariance matrix and plot a heatmap of the covariance matrix of selected features using **Seaborn** package
- b) Plot the correlation **heatmap** of the correlation matrix of selected features using **Seaborn** package
- c) Plot scatterplots of 5 highly-correlated pairs of features and 5 non correlated pairs of features.
- d) Use **pandas.plotting.scatter_matrix** for the selected features.

Note:

pandas.plotting.scatter_matrix is a function within the pandas library that generates a matrix of scatter plots and histograms (or kernel density estimates) for a given DataFrame. This visualization tool is particularly useful for exploring relationships and distributions among multiple numerical variables in a dataset.

- **Scatter Plots:**

For every unique pair of numerical columns in the input DataFrame, scatter_matrix creates a scatter plot, allowing for the visual inspection of **bivariate relationships and potential correlations**.

- **Histograms/KDEs:**

Along the diagonal of the matrix, the function displays either histograms or Kernel Density Estimation (KDE) plots for each individual numerical variable. These plots illustrate the **distribution of each variable**.

Question 3:

Use '**diabetes.csv**'.

Rank the features by importance using Random Forest Classifier.

Plot a bar chart of feature importance.

Reference:

https://scikit-learn.org/stable/auto_examples/ensemble/plot_forest_importances.html

Hint:

```
clf = RandomForestClassifier(n_estimators=100, random_state=42)
```

```
...
```

```
feature_importances = clf.feature_importances_
```